

The Spark + tech stack.

A hardened clean architecture for privacy, deterministic reliability, and just-in-time personalization.

Five layers, one offer.

Edge-first AI keeps PII on-device. A deterministic backend gates entitlement. A learning graph explains every decision.

01 · FRONTEND

Privacy-first PWA

- Cross-platform React PWA
- Edge AI via Gemma 3n on Google AI Edge
- H3 grid quantization (~50m)

React

Gemma 3n

H3

02 · BACKEND

Deterministic FastAPI

- Gemini Flash for GenUI framing
- Rules-first decision engine
- Domain-Driven Design isolation

FastAPI

Gemini Flash

DDD

03 · GRAPH

Self-learning brain

- Weighted preference decay
- Machine-readable audit trace
- Continuity-id pseudonyms

Neo4j

Cypher

04 · INFRA

Pipeline & guardrails

- Fluent Bit + Lua ingestion
- SQLite audit & idempotency
- Ruff, Pyright, AST guards

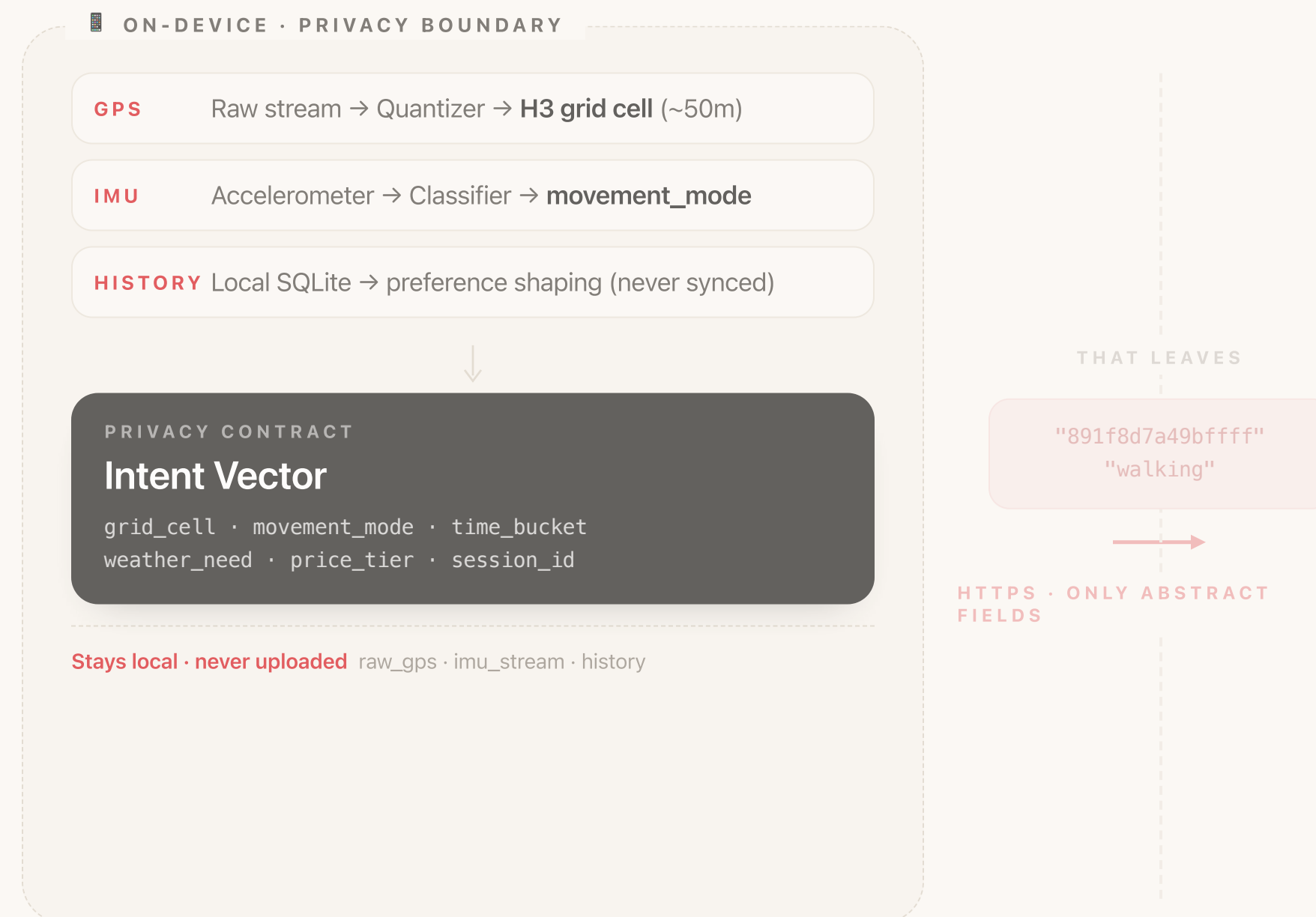
Fluent Bit

SQLite

uv

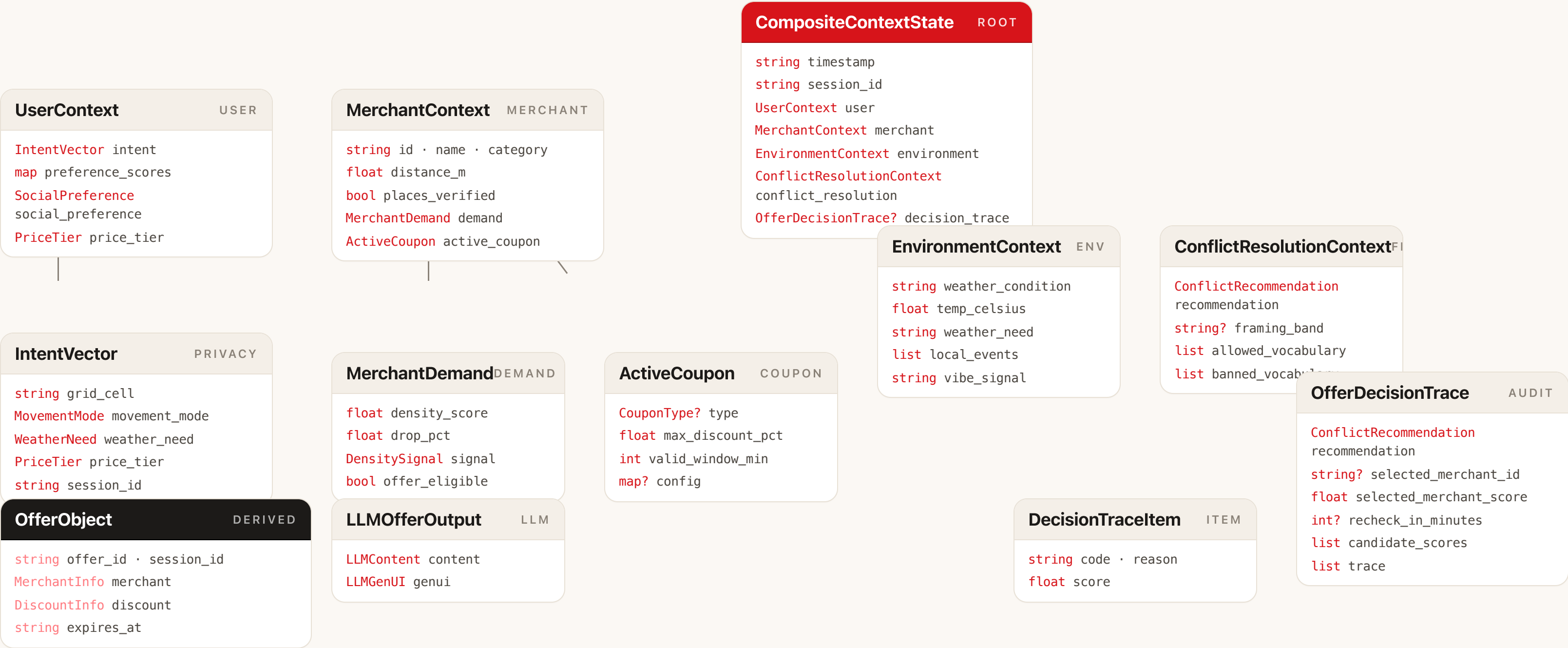
Raw data stays on device. Only an **Intent Vector** crosses.

On-device quantizers abstract every signal before it leaves. The cloud folds it together with four external feeds into one Composite Context State.



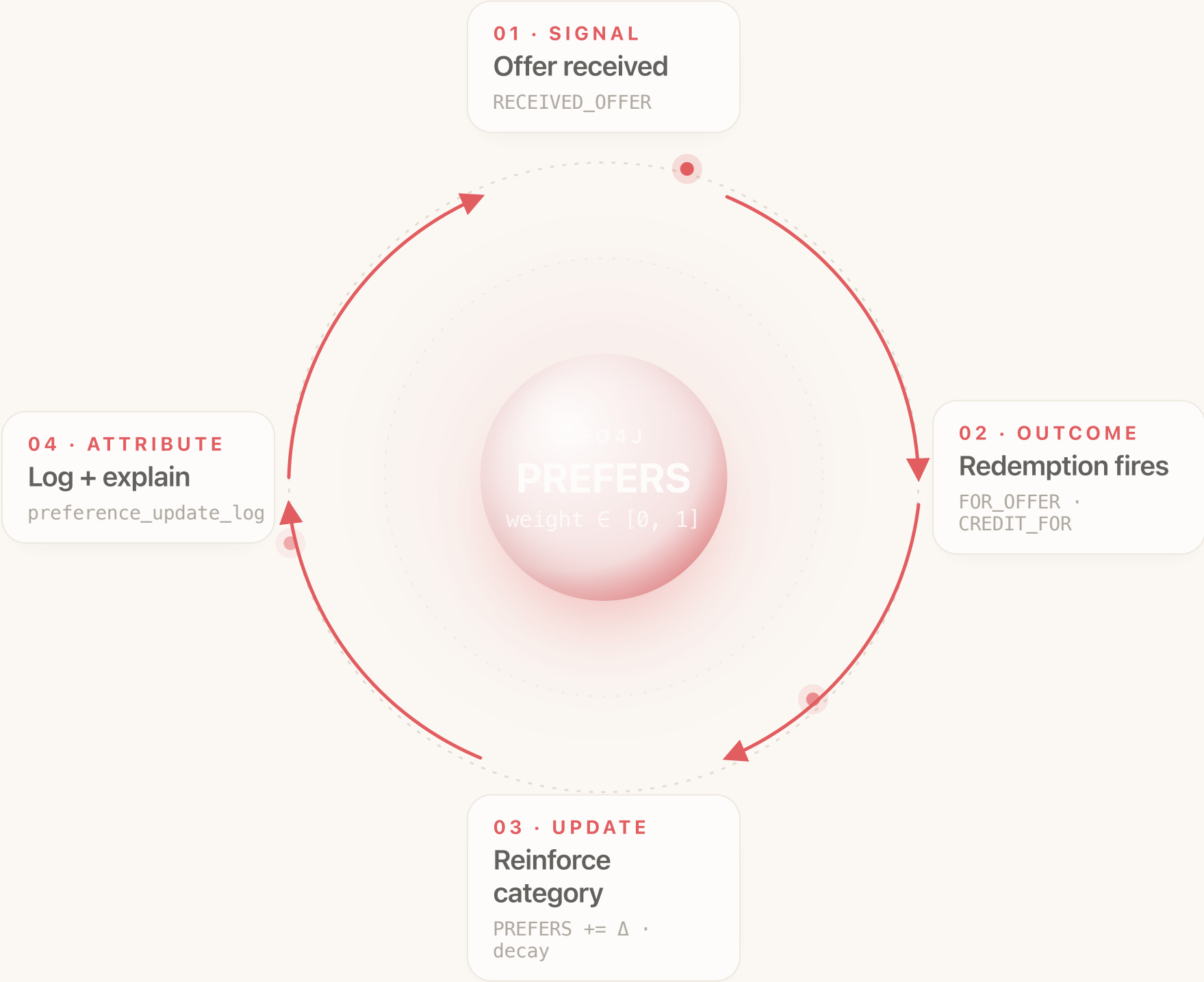
Eleven typed contracts compose every offer.

CompositeContextState aggregates user, merchant, environment and decision trace.
OfferObject is derived, never invented.



The brain that learns from every redemption.

Outcomes flow back into **PREFERS** weights — bounded, idempotent, and fully audit-traceable.



Edges that learn.

Every interaction is a typed edge with a bounded delta and provenance.

UserSession	RECEIVED_OFFER	Offer
Offer	AT_MERCHANT	Merchant
Offer	GENERATED_IN	Context
Redemption	FOR_OFFER	Offer
WalletEvent	CREDIT_FOR	Redemption
UserSession	PREFERS · WEIGHT	Category